Shreyas Gupta

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Education

- **Indian Institute of Information Technology, Allahabad** — Prayagraj, India
  Bachelor of Technology - Information Technology; GPA: 8.93 — *July 2019 - June 2023*

Experience

- **Sprinklr** — Gurugram, Haryana
  - ***Senior Product Engineer - Backend*** — *May 2025 - Present*
    - Led the **distributed architecture refactor** of our call-routing engine:
      - Redesigned a master-slave architecture to a **Redis**-backed **distributed system**, with **Zookeeper** for **distributed locking** to ensure correctness.
      - Achieved **horizontal scalability** resulting in over **40% infra cost reduction** for the engine.
      - **Reduced** critical service **downtimes** to **zero**, severely improving on **fault tolerance** and **high availability**.
      - Ensured careful balance of **high availability, reliability, correctness, and speed**
  - ***Product Engineer - Backend*** — *Jan 2023 - Apr 2025*
    - Refactored a continuously running voice call router into an **event-driven** model, **improving CPU utilization** of the call-routing engine by **up to 33%**.
    - Benchmarked multiple deep-cloning libraries for speed, accuracy, and CPU usage, and created a custom deep-cloning utility that **improved performance by 6-8x** for mission-critical workloads.
    - Designed and implemented a base-level interceptor to **simulate service outages** (e.g. Elasticsearch outage, Redis outage, etc) in a microservice, effectively measuring and helping **reduce the blast radius** across various modules to make the system more **fault tolerant**.
    - Developed a **Kafka-powered** monitoring framework for our call-routing engine, aggregating real-time performance and operational metrics plotted through **Kibana** dashboards, helping **minimizing downtime** through **faster issue identification**.
    - Eliminated Elasticsearch dependency from a mission-critical workflow by implementing **asynchronous data sync** and **Kafka-based retries**, **reducing the impact of Elasticsearch outages** on critical flows.
  - ***Product Engineer Intern - Backend*** — *May 2022 - July 2022*
    - Wrote a **Spring Boot** service to fetch and analyze historical case routing state of a customer care agent. Worked with **MongoDB** and **Elastic Search**.
    - Wrote a **Spring Boot** service to **automate testing** for chatbots. Worked with **MQTT protocol** for communication with the bot.
- **Logward** — Remote
  - ***Backend Developer Intern*** — *May 2021 - July 2021*
    - Wrote a **monitoring engine** to monitor mission critical data pipelines helping find points of failure faster and automating retries.
    - Wrote the micro-service in **Typescript** and also wrote **.NET and Python libraries** to integrate with services involved in pipeline. Worked with many AWS services like SQS and S3. Also had exposure to **Kubernetes**.

Skills Summary

- **Languages**: Java (Spring Boot), Python, Node.js (ExpressJS), JavaScript / TypeScript, C++, SQL
- **DBs and Tools**: MongoDB, Redis, Kafka, Elastic Search, MySQL, Git
- **Cloud/Infra**: Kubernetes, Docker

Projects

- **PuzzleDash**
  https://puzzledash.shreyasgupta.in — *Jan 2022*
  - A chess minigame where you solve a series of chess puzzles with increasing difficulty in limited time.
  - Hosts **50+ monthly active users**.
  - Worked with Socket.io for **realtime communication** and Node.js for the backend server.
- **Impedia**
  https://impedia.shreyasgupta.in — *April 2023*
  - Co-authored the backend of Impedia, a platform to effectively communicate grievances or requests within your organization.
  - Onboarded **4 organizations** with a total of **100+ users** in the prototype stage.
  - Developed using Nodejs(Express) with MongoDB. Implemented authentication using JWTs for multiple user types.